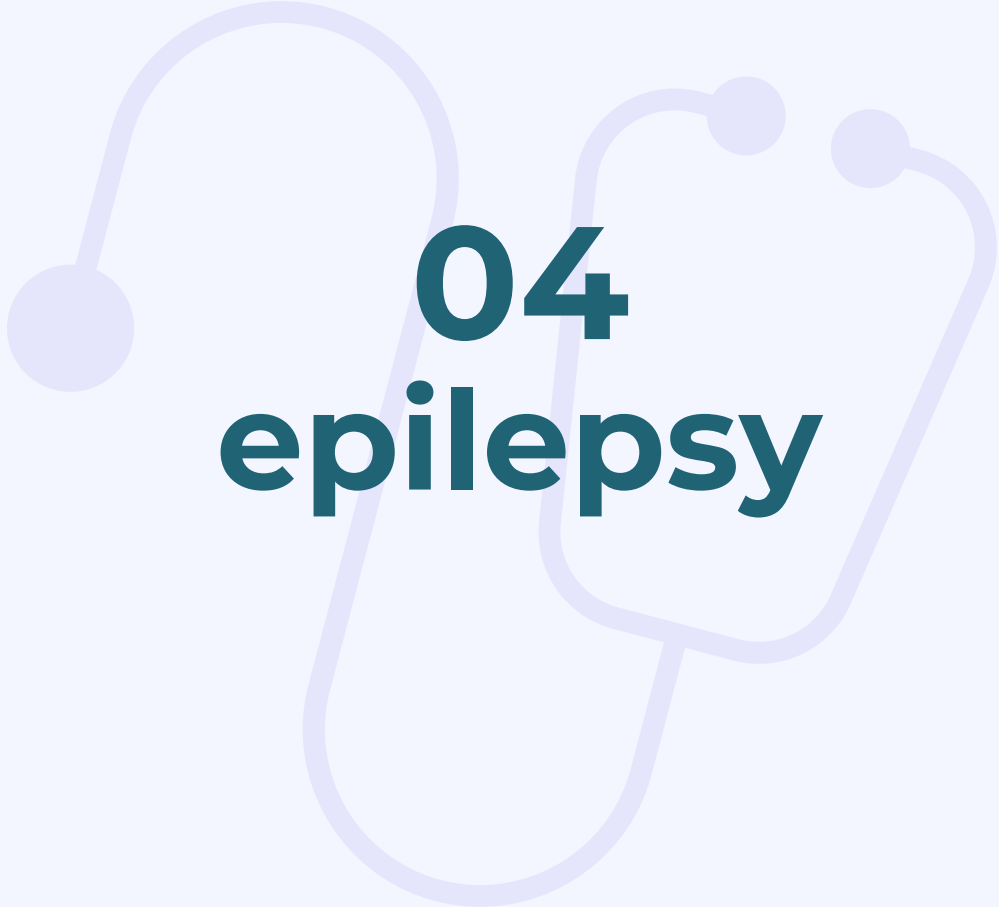


# CNS

By Dr.Riffat Ibrahim Riffat

A large, light purple outline of a human head in profile, facing right. A stethoscope is draped around the neck and over the head. The background is a light blue gradient with various medical icons: a clipboard with a checklist, a green pill bottle with a white cross, a red heart with a white cross, and a red and white pill bottle. There are also several small white plus signs and a trail of small blue dots.

# 04 epilepsy

# Introduction

- epilepsy :
  - is a disease characterized by recurrent episodes of seizuers.
- Seizuers:
  - Are transient alterations of brain functions (motor, consciousness, visual, sensation, smell) due to abnormal neuronal discharge.
  - A single seizure does not qualify a patient to be diagnosed with epilepsy.
- Convulsions:
  - Involuntary often violent, spasms for skeletal muscles, could be related to seizures or not.



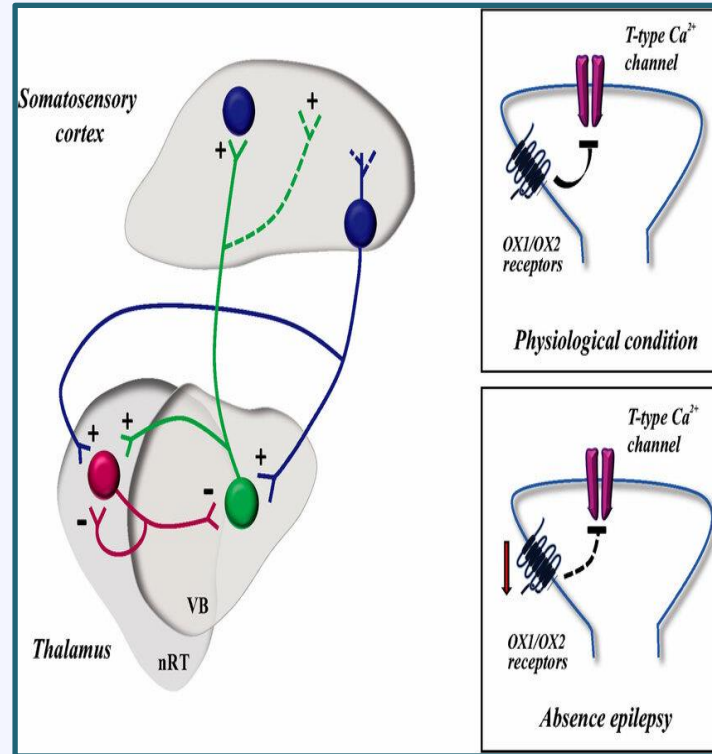
# Patho physiology

All types of epilepsy can be explained by the following theory except absence seizures:

- An imbalance between glutamate and which activates NMDA receptors (excitatory) and the inhibitory GABA neurotransmitter leading to abnormal neuron activation.

In absence seizures :

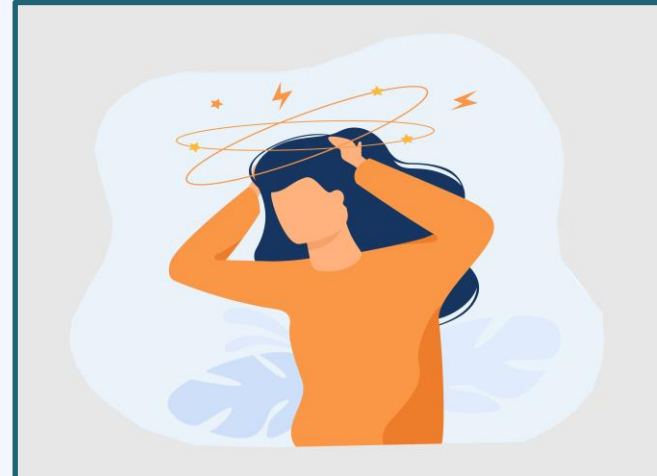
- Activation of t-type calcium channels in the relay neuron during the day causing a sleep like state.
- + This type is common in children



# Causes

• Causes of epilepsy maybe be:

- Genetic causes : affecting GABA production or receptors.
- Acquired:
  - Infections , eg: meningitis .
  - Head trauma .
  - Stroke (Brain bleeding ) .



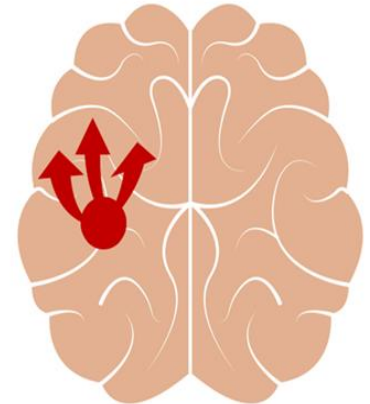
# Classification of Epilepsy

Focal-onset epilepsy : localized in one area and in one hemisphere, the foci size will affect the symptoms (larger foci produces more noticeable symptoms)

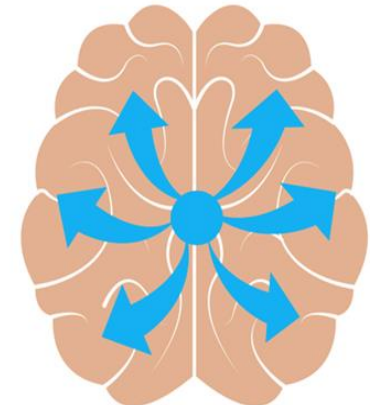
- Motor cortex: simple limb jerking or convulsions
- Visual area: visual abnormalities.
- Temporal cortex: auditory abnormalities.

Types :

- Simple focal seizures (With preserved awareness).
- Complex Focal seizures (impairment of awareness or loss of awareness)
- Focal seizures with secondary generalization .



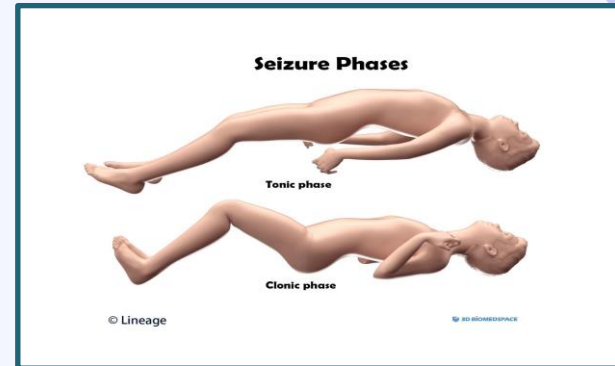
**FOCAL**



**GENERALIZED**

Generalized Onset seizures affecting both hemispheres of the brain:

- Absence seizures (Petit mal or little illness), a child can have over 100 mini episodes daily lasting 3-5 seconds.
- Tonic-Clonic seizures (Grand mal or Great illness).
- Myoclonic seizures with jerking movement.
- Loss of consciousness is constant with all three types of generalized seizures.



#### GENERALIZED SEIZURE: OTHER TYPES



##### Myoclonic Seizure

Non-rhythmic jerks resulting from involuntary muscle twitching that normally target the upper extremities, and after an episode, patients may describe the perceived sensation as momentary electrical shocks.

# First generation drugs

- Members include:
  - Phenytoin
  - Carbamazepine
  - Valproic acid
  - Ethosuxiamide
  - phenobarbitone (very limited application)
  - Benzodiazepens. (limited to status epileptics I.V only and febrile seizures I.V or rectal suppositories)
- They mainly act on the inhibitory pathway.
- Adverse effect profile is very high with these agents, except ethosuxiamide.



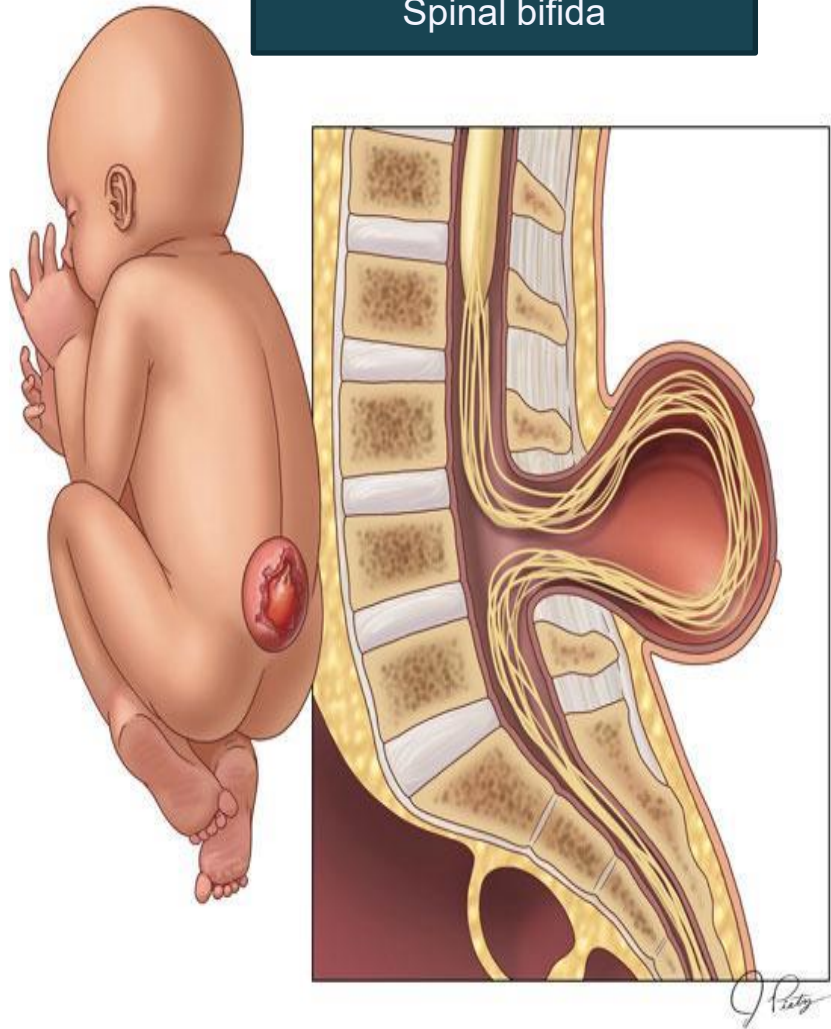
- Phenytoin or diphenyl hydantoin.
- MOA: Blocking sodium channels in Brain and heart.
- Therapeutic uses :
  - Treatment of Tonic-Clonic Seizures, other types of seiziures can be treated by phenytoin, however, newer agents are more superior and are safer.
  - Anti-arrhythmic drug .



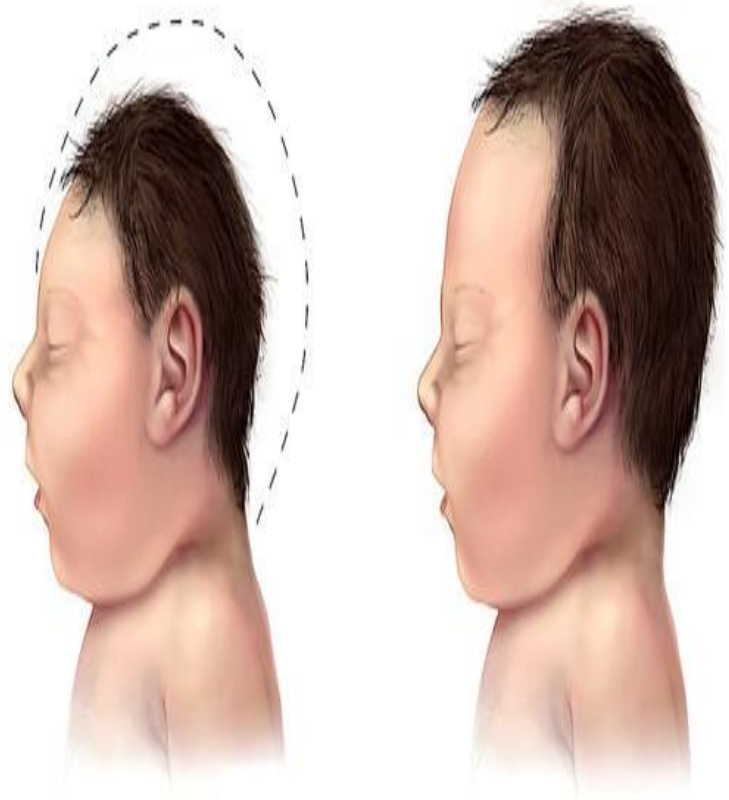
## Adverse effects :

- The following adverse effects are common with all 1<sup>st</sup> generation epileptic drugs:
  - CNS (Nystagmus , Diplopia and Ataxia ).
  - Liver (microsomal enzyme induction )
  - Blood ( Megaloblastic anemia) due to decrease in folic acid caused by microsomal enzyme induction
  - Teratogenicity
    - It is about 3-4%, this effect is folic acid related, and supplementation is required with continuing the drug. Breast feeding is not an issue.
    - Craniofacial abnormality (Fetal hydantoin syndrome)
    - Neural tube defect (Spinal bifida )

Spinal bifida



Fetal hydantoin syndrome



## Adverse effects :

- The following adverse effects are specific to phenytoin:

- Gingival hyperplasia.



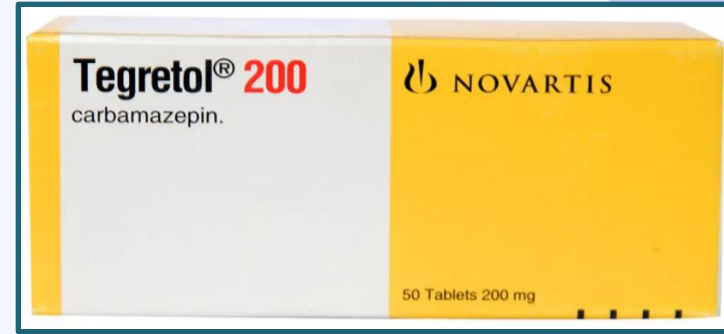
- Hirsutism.

- Osteomalacia, due to microsomal induction of vit. D metabolism.

- Lymphadenopathy.



- Carbamazepine:
- MOA: Block Sodium channel in brain only
- Therapeutic Uses:
  - First choice in focal seizures.
  - First choice in trigeminal neuralgia.
  - Second choice in tonic-clonic seizures.
- Adverse effects:
  - Increase ADH secretion. (edema with hyponatremia)
  - Carbamazepine is characterized by auto-induction of its own metabolism.



- Valproic acid:

- MOA :

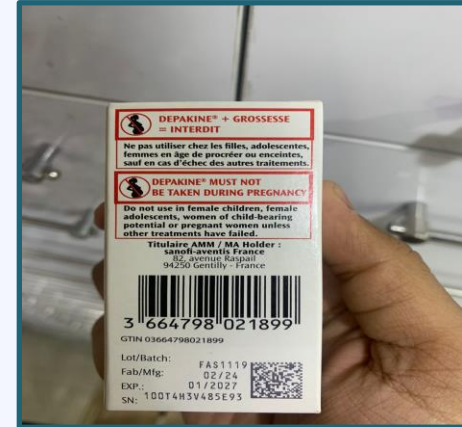
- increase GABA inhibitory transmitter ,through inhibiting GABA transaminase that inhibits GABA metabolism.
- Block Sodium channels in brain .
- Block T-type calcium channels .

- Therapeutic Uses:

- Treatment of all types of epilepsy.

- Adverse effects:

- Liver : microsomal enzyme inhibitor.
- Neutropenia .
- Alopecia
- Pancreatic and hepatic dysfunction (Rare but leads to death).
- Valproic acid is strictly prohibited in pregnancy, as it has the ability to cause major abnormalities. (the drug comes with a black box warning)



- Ethosuximide:
- MOA: Selective T-type calcium channel blocker.
- Therapeutic uses: treatment of Absence seizures Only.
- Side effects:
  - Ethosuximide is the most safe anti-epileptic drug for absence seizures treatment.
  - GIT disturbance and Headache.





# second generation drugs



- Members include:
  - Lamotrigine
  - Topiramate
  - Levetiracetam
  - Gabapentin and pregabalin
- They mainly act on the excitatory pathway.
- Adverse effect profile is much lower with these agents than 1<sup>st</sup> generation.





- Lamotrigine:
- MOA:
  - Blocking Sodium Channels
  - Inhibition of Glutamate.



- Therapeutic Uses:
  - Treats All types of Epilepsy Generalized, Focal and, even absence seizures.
  - Best choice in pregnancy
- Side effects:
  - Steven Johnson syndrome, Which increases with the combination of Lamotrigine and Valproic acid.

- Topiramate:
- MOA:
  - Block Sodium channels.
  - Potentiates GABA action.
  - Block AMBA receptors. (similar to NMDA receptor)
- Therapeutic uses:
  - Treatment of all types of epilepsy.
  - Migraine prophylaxis by an unknown mechanism
- Side effects:
  - Increase intraocular pressure (Glaucoma).
  - Renal stones.
  - Weight loss. (can be used for weight reduction, but it is still under clinical investigations)



- Levetiracetam:

- MOA:

- Binding Selectively to the vesicular protein SV2A and inhibition of Excitatory transmitters release.

- Therapeutic Uses:

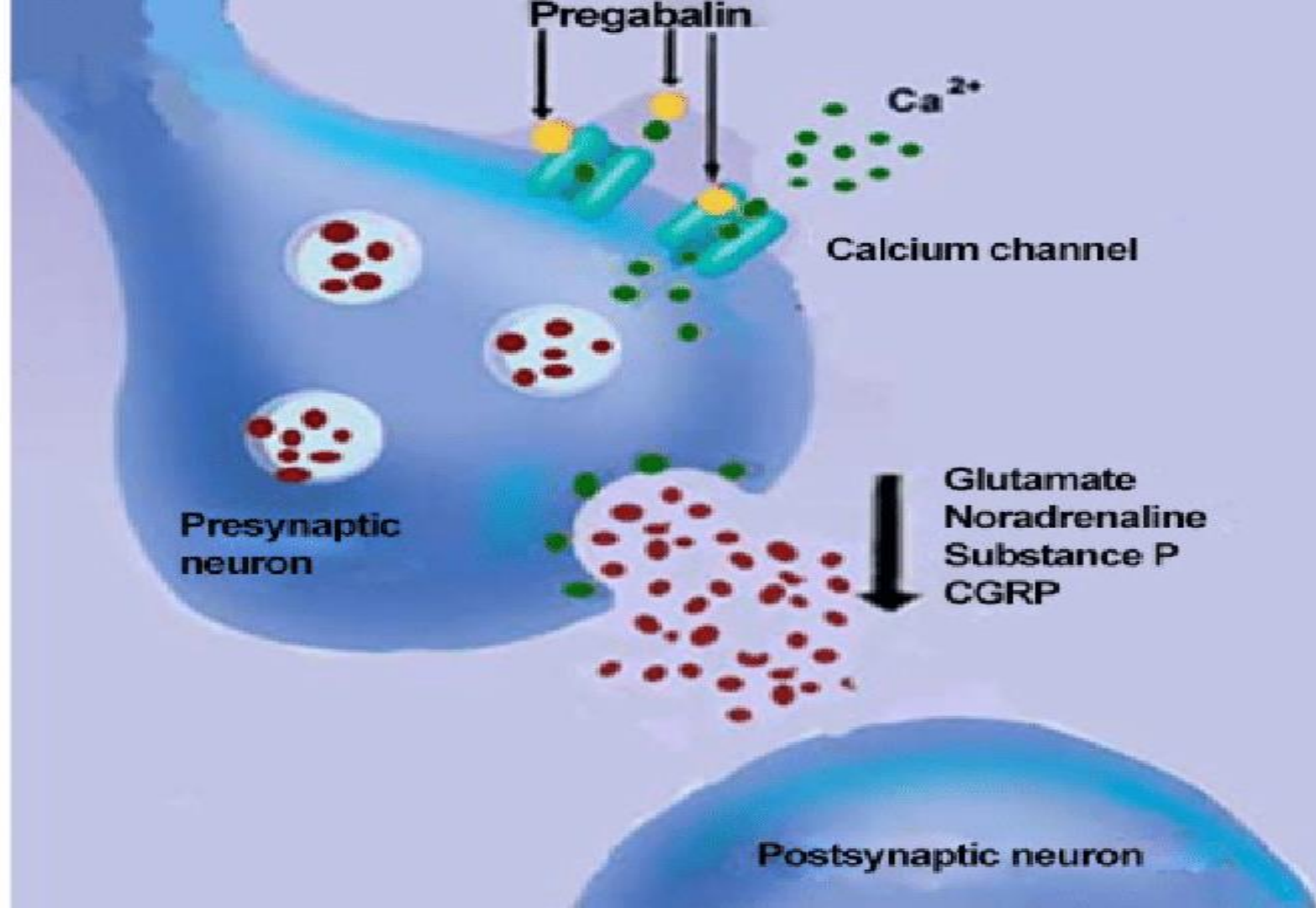
- Treatment of all types of epilepsy.

- Levetiracetam has very low side effects profile and is very safe in pregnancy.



- Gabapentin and pregabalin:
- MOA:
  - Binding to pre-synaptic P/Q calcium channels, preventing calcium influx and reducing glutamate release.
- Therapeutic Uses:
  - Adjuvant to other antiepileptic drugs.
  - Ideal drugs for neuropathy
- Adverse effects:
  - Sedation
  - Drug dependence and addiction, pregabalin is more likely to cause addiction than Gabapentin.





# Cases

- A child is experiencing absence seizures that interrupt his ability to pay attention during school and activities.
- Which of the following therapies would be most appropriate for this patient?
  - Ethosuximide.
  - Carbamazepine.
  - Diazepam.
  - Carbamazepine plus primidone.
  - Watchful waiting.

# Answers

- Correct answer = A. The patient has had many seizures that interrupt his ability to pay attention during school and activities, so therapy is justified. Monotherapy with primary agents is preferred for most patients. The advantages of monotherapy include reduced frequency of adverse effects, fewer interactions between antiepileptic drugs, lower cost, and improved compliance. Carbamazepine and diazepam are not indicated for absence seizures.

# Cases

- A 25-year-old woman with myoclonic seizures is well controlled on valproate. She indicates that she is interested in becoming pregnant in the next year. With respect to her antiepilepsy medication, which of the following should be considered?

- A. Leave her on her current therapy.
- B. Consider switching to lamotrigine.
- C. Consider adding a second antiepilepsy medication.
- D. Decrease her valproate dose.



# Answers

- Correct answer = B. Valproate is a poor choice in women of child-bearing age. A review of the medication history of this patient is warranted. If she has not tried any other antiepilepsy medication, then consideration of another anti-epilepsy medication may be beneficial. Studies show that valproate taken during pregnancy can have a detrimental effect on cognitive abilities in children.

# Cases

- A 25-year-old woman with myoclonic seizures is well controlled on valproate. She indicates that she is interested in becoming pregnant in the next year. With respect to her antiepilepsy medication, which of the following should be considered?

- A. Leave her on her current therapy.
- B. Consider switching to lamotrigine.
- C. Consider adding a second antiepilepsy medication.
- D. Decrease her valproate dose.

# Answers

- Correct answer = B. Valproate is a poor choice in women of child-bearing age. A review of the medication history of this patient is warranted. If she has not tried any other antiepilepsy medication, then consideration of another anti-epilepsy medication may be beneficial. Studies show that valproate taken during pregnancy can have a detrimental effect on cognitive abilities in children.